

# GxP / ATO / eSignature

## Complete End-to-End Setup Guide

From Add-on Installation to PO Registration → ATO OData V4 → eSignature → Reauthentication → BTP Comm Arrangement

SAP S/4HANA Public Cloud | BTP ATO SaaS | Scope Item 84P | 2026-06-26

### Complete Setup Sequence

| Phase    | Title                                             | System               | When                |
|----------|---------------------------------------------------|----------------------|---------------------|
| Phase 1  | Install Life Science Add-on in S4                 | S4 — Basis           | One-time            |
| Phase 2  | Activate Scope Item 84P — GxP Framework           | S4 — SPRO            | One-time            |
| Phase 3  | Register Purchase Order Object in VC_SRA_SOT      | S4 — SPRO            | One-time per LOB    |
| Phase 4  | Implement PO Relevancy Check Callback Class       | S4 — SE24            | Dev task            |
| Phase 5  | Enable ATO Fields in PO OData V4 API              | S4 — /IWBEP/V4_ADMIN | CE2608 delivered    |
| Phase 6  | Configure ATO Extension Tables & SEGW             | S4 — SE11 + SEGW     | POC — CGTO_S4HC_POC |
| Phase 7  | Register ATO Fields in Change Document Object     | S4 — SCDO            | CE2608 / ATO team   |
| Phase 8  | Set Up BTP eSignature SaaS Communication          | S4 + BTP             | One-time            |
| Phase 9  | Configure BTP — Register PO Object & Reason Codes | BTP                  | One-time per LOB    |
| Phase 10 | Integrate REQUEST_REAUTH in ME21N Save Flow       | S4 — SE24 / BAdI     | Dev task — 2702     |
| Phase 11 | Set Up ATO SaaS — Event Mesh & iFlow              | BTP                  | One-time            |
| Phase 12 | End-to-End Validation                             | All systems          | After all phases    |

Reading guide: Each phase has numbered steps. Complete them in order — phases are dependent on each other. Phase 5 (OData V4 extensibility) was delivered in CE2608 and may already be done. Always verify before repeating.

# 1

## Install Life Science Add-on in S4

*Delivers /ATO/ extension tables, OData extensions, Change Document registration*

The Life Science Add-on (LS\_ADD\_ON) is delivered by SAP Cloud Operations. It installs all /ATO/ namespace objects, extension tables, and wires up the PO OData API. This is NOT a manual step — it requires SAP to apply the transport.

### 1

#### Request Add-on Installation

Raise a request with SAP Cloud Operations to install LS\_ADD\_ON / ATO\_S4HC on the target system. Life Science Add-on = ATO + SI + CSAI + CSM + BRH combined package. Specify: target system ID, release version, required by date.

### 2

#### SAP applies add-on transport

SAP Cloud Operations applies the transport. Delivered objects: /ATO/S4\_PO\_HEADER\_EXT — PO Header extension table (FK → EKKO) /ATO/S4\_PO\_ITEM\_EXT — PO Item extension table (FK → EKPO) Industry Object Extension nodes on EKKO and EKPO PurchaseOrder OData V4 API extended with TherapyType, ClinicalStudy, ProtocolRef, PatientRef Change Document EINKBELEG extended to include /ATO/S4\_PO\_HEADER\_EXT Package CGTO\_S4HC\_POC with all objects

### 3

#### Verify: SAINT

Transaction: SAINT → Status tab Expected: LS\_ADD\_ON or ATO\_S4HC → Status = INSTALLED with version

### 4

#### Verify: SE11 extension tables

SE11 → /ATO/S4\_PO\_HEADER\_EXT → Status = Active SE11 → /ATO/S4\_PO\_ITEM\_EXT → Status = Active Both must exist before any further configuration

### 5

#### Verify: SE03 namespace

SE03 → Namespace → /ATO/ → Status = ACTIVE

### 6

#### Verify: SCDO Change Document

SCDO → EINKBELEG → Table Names tab /ATO/S4\_PO\_HEADER\_EXT must be in the list → confirms CE2608 change doc work applied

■ Phase 1 complete when SAINT=INSTALLED, both SE11 tables Active, SCDO has /ATO/ table.

## 2

### Activate Scope Item 84P — GxP Framework

*S4 + BTP activation — switches on Reauthentication Framework and provisions BTP tenant*

#### 1

##### Access SAP Cloud Solution Configuration

In S4HC Admin Launchpad → Solution Configuration → Scope OR via SAP Central Business Configuration (CBC)

#### 2

##### Search for Scope Item 84P

Search field: enter '84P' or 'GxP' Scope item name: GxP Compliance Select it and review the description

#### 3

##### Activate 84P

Click Activate / Include in Scope System will prompt for transport request → assign to your project transport Activation triggers: • VC\_SRA\_SOT S-table entries created in S4 • SPRO → GxP Compliance menu becomes available • BTP eSignature SaaS tenant provisioned automatically

#### 4

##### Verify S4 activation — SPRO

SPRO → search for 'GxP' or 'Reauthentication' GxP Compliance customising node should appear in SPRO tree

#### 5

##### Verify BTP provisioning

BTP Cockpit → your subaccount → Instances and Subscriptions eSignature SaaS service should appear as provisioned

#### 6

##### Test CHECK\_ESIG\_ACTIVE

After activation → CHECK\_ESIG\_ACTIVE should return ABAP\_TRUE If FALSE: verify Communication Arrangement (Phase 8) is also configured

■ 84P activation is a system-wide change. Activating it enables GxP compliance mode for ALL registered business objects. Ensure all LOB teams are ready before activating in production. Always activate in a test/sandbox system first.

### 3

## Register Purchase Order Object in VC\_SRA\_SOT

*SPRO configuration — tells the framework which PO fields require reauthentication*

#### 1

##### Open SPRO GxP node

Transaction: SPRO → search 'GxP Compliance' or 'Reauthentication' Navigate to: GxP Compliance → Reauthentication → Define Reauthentication Objects

#### 2

##### Create PURCHASE\_ORDER entry

Click New Entry Reauth Object: PURCHASE\_ORDER Description: Purchase Order GxP Reauthentication Reauth Class: ZCL\_PO\_REAUTH\_CHECK (callback class — implement in Phase 4) Auth Class: ZCL\_PO\_AUTH\_CHECK (authorization class — implement in Phase 4) Save

#### 3

##### Define Node Types for PO

Select PURCHASE\_ORDER entry → Navigate to Node Types Create node 1: Node Name: PO\_HEADER\_EXT Table Name: /ATO/S4\_PO\_HEADER\_EXT Description: ATO Header Extension Fields Create node 2: Node Name: PO\_ITEM\_EXT Table Name: /ATO/S4\_PO\_ITEM\_EXT Description: ATO Item Extension Fields

#### 4

##### Define Regulated Fields per Node

Select PO\_HEADER\_EXT node → Navigate to Field Configuration Add regulated fields (one row each): Field Name: THERAPY\_TYPE Description: Therapy Type Field Name: CLINICAL\_STUDY Description: Clinical Study Reference Field Name: PROTOCOL\_REF Description: Protocol Reference Select PO\_ITEM\_EXT node → Field Configuration Field Name: THERAPY\_TYPE Field Name: PATIENT\_REF

#### 5

##### Save and transport

Save all configuration → assign to transport request This content will be delivered as part of 84P transport to other systems

#### 6

##### Verify via SE16N

SE16N → VC\_SRA\_SOT Confirm rows exist for PURCHASE\_ORDER with correct class names and nodes

The Relevancy Check callback class (ZCL\_PO\_REAUTH\_CHECK) and authorization class (ZCL\_PO\_AUTH\_CHECK) must be registered here BEFORE implementing them in Phase 4. The class names must match exactly — VC\_SRA\_SOT looks them up by name at runtime.

## 4

## Implement PO Relevancy Check Callback Class

SE24 — ABAP class that decides whether reauthentication is required

## 1

## Create ZCL\_PO\_REAUTH\_CHECK in SE24

SE24 → Class → ZCL\_PO\_REAUTH\_CHECK → Create Class Type: Regular Class Final: Yes Description: PO Reauthentication Relevancy Check

## 2

## Add interface IF\_REAUTH\_RELEVANCY\_CHECK

Interfaces tab → Add: IF\_REAUTH\_RELEVANCY\_CHECK This interface provides the IS\_REAUTHENTICATION\_RELEVANT method signature

## 3

## Implement IS\_REAUTHENTICATION\_RELEVANT

Methods tab → double-click IS\_REAUTHENTICATION\_RELEVANT → implement:

```
METHOD if_reauth_relevancy_check-is_reauthentication_relevant.

    " Get regulated field list from S-table via framework utility

    DATA: lt_reg_fields TYPE TABLE OF vc_sra_sot_fields,
           lv_relevant   TYPE abap_bool VALUE abap_false.

    " Call framework utility to read configured regulated fields

    " for PURCHASE_ORDER object from VC_SRA_SOT

    " (Framework provides this utility – LOB does not query VC_SRA_SOT directly)

    " Check if any of the regulated fields changed in this save

    " For ATO fields on PO header extension:

    IF im_changed_fields-therapy_type IS NOT INITIAL OR
       im_changed_fields-clinical_study IS NOT INITIAL OR
       im_changed_fields-protocol_ref IS NOT INITIAL OR
       im_changed_fields-patient_ref IS NOT INITIAL.

        lv_relevant = abap_true.
    ENDIF.

    " Return result

    re_relevant = lv_relevant.

ENDMETHOD.
```

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**4****Create ZCL\_PO\_AUTH\_CHECK**

SE24 → ZCL\_PO\_AUTH\_CHECK → Create Implements authorization check interface Logic: check if current user has authority to perform this regulated change For CRUD operations: check CREATE/CHANGE authority on purchase order object

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**5****Activate both classes**

SE24 → ZCL\_PO\_REAUTH\_CHECK → Activate (Ctrl+F3) SE24 → ZCL\_PO\_AUTH\_CHECK → Activate (Ctrl+F3)

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**6****Test with ABAP Unit**

SE24 → Local Test Classes tab → write unit tests: Test 1: ATO field changed → method returns ABAP\_TRUE Test 2: Non-ATO field changed → method returns ABAP\_FALSE Test 3: No fields changed → method returns ABAP\_FALSE

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■ Phase 4 complete when both classes are Active and unit tests pass.

## 5

## Enable ATO Fields in PO OData V4 API

CE2608 deliverable — internal extensibility already enabled by PO standard team

- CE2608 STATUS: This phase was delivered in CE2608 by the PO standard team (Nils/Jiss). The OData V4 PurchaseOrder API internal extensibility is already enabled. Verify it is in place before proceeding. If already enabled — skip to Phase 6.

## 1

### Verify OData V4 API extensibility

Transaction: /IWBEP/V4\_ADMIN Search for service: API\_PURCHASEORDER\_2 Check: Internal Extensibility = Enabled If Enabled → proceed to Phase 6. If NOT → follow steps below.

## 2

### Enable extensibility (if not already done)

This requires the PO standard team (Nils Hartmann / Jiss) to apply the enablement. Contact: nils.hartmann@sap.com Reference: CE2608 — 'OData V4 API enabled for internal extensibility (without the PaaS API extension)'

## 3

### Verify ATO extension visible in \$metadata

Gateway Client → GET /sap/opu/odata4/sap/api\_purchaseorder\_2/srvd\_a2x/sap/purchaseorder/0001/\$metadata Search response XML for: TherapyType, ClinicalStudy, ProtocolRef, PatientRef These properties must appear in the PurchaseOrder entity type

## 4

### Test OData call with ATO fields

Gateway Client → POST to PurchaseOrder API Include ATO fields in request body: "to\_PurchaseOrderHeaderExtension": { "TherapyType": "CELL\_GENE", "ClinicalStudy": "CS-TEST-001" } Verify response includes ATO fields and /ATO/S4\_PO\_HEADER\_EXT row created

## 6

## Configure ATO Extension Tables in SEGW

*Register /ATO/ tables as separate Business Object — Industry Object Extension pattern*

Architecture decision (confirmed in meetings): ATO fields are stored in SEPARATE Business Objects (/ATO/S4\_PO\_HEADER\_EXT and /ATO/S4\_PO\_ITEM\_EXT) — NOT appended directly to EKKO/EKPO. This enables third-level extension which native EKKO append cannot support.

## 1

### Verify tables exist and are active

SE11 → /ATO/S4\_PO\_HEADER\_EXT → Active SE11 → /ATO/S4\_PO\_ITEM\_EXT → Active Fields: EBELN (FK), THERAPY\_TYPE, CLINICAL\_STUDY, PROTOCOL\_REF (header) EBELN+EBELP (FK), THERAPY\_TYPE, PATIENT\_REF (item)

## 2

### Verify Industry Object Extension node is registered

The add-on (Phase 1) registers the extension nodes on EKKO and EKPO. Verify: EKKO has an Industry Object Extension node pointing to /ATO/S4\_PO\_HEADER\_EXT Check via: SE11 → EKKO → Display → Enhancement Categories / Extension nodes

## 3

### Verify SEGW service exposes ATO fields

SEGW → open PurchaseOrder OData service (if custom SEGW service exists for ATO) OR verify via /IWBEP/V4\_ADMIN that TherapyType etc. are in the entity type The add-on delivers the OData extension automatically

## 4

### Verify data in extension tables after PO creation

Create a test PO via ATO or ME21N with ATO fields populated SE16N → /ATO/S4\_PO\_HEADER\_EXT → filter EBELN = Confirm row exists with correct THERAPY\_TYPE, CLINICAL\_STUDY, PROTOCOL\_REF

## 5

### Verify ME23N shows ATO tab

ME23N → open PO → look for 'Advanced Therapy Information' tab on header Tab should show Therapy Type, Clinical Study, Protocol Ref fields This tab is delivered by the add-on — no custom Fiori development needed

■ Phase 6 complete when /ATO/ tables have data and ME23N shows ATO tab.



## 7

## Register ATO Fields in Change Document Object (EINKBELEG)

*Ensures ATO field changes are recorded in CDHDR/CDPOS — GxP audit trail requirement*

Status: ATO team (Allen Yuan) completed a POC of this registration. Approach: extend EINKBELEG to include /ATO/S4\_PO\_HEADER\_EXT + adjust function module. Open issue: ATO change rows appear in CDPOS but description is blank in ME23N change log screen.

### 1 Verify EINKBELEG already includes ATO table

SCDO → EINKBELEG → Table Names tab If /ATO/S4\_PO\_HEADER\_EXT is listed → Phase 1 add-on applied this. Skip to step 4. If NOT listed → proceed with steps 2-3

### 2 Extend EINKBELEG with /ATO/S4\_PO\_HEADER\_EXT

SCDO → EINKBELEG → Change Table Names tab → Add row: Table Name: /ATO/S4\_PO\_HEADER\_EXT Documentation Key: ATO Header Extension Save → Activate → assign transport NOTE: This requires contact with ATO team (Allen Yuan / Loring Wu) to confirm approach

### 3 Adjust function module for ATO change recording

ATO team confirmed: a function module was adjusted to SEPARATELY call the change document recording for ATO fields when a PO with ATO data is saved. This is called in the PO BAdI body (ME\_PROCESS\_PO\_CUST or equivalent). Contact: Allen Yuan (ATO dev team) for the exact function module name

### 4 Add implicit enhancement for description text

ATO team added an implicit enhancement in the standard programme to display description text for /ATO/ table fields in the ME22N/ME23N change log screen. Acceptability of this approach to be confirmed with PO standard team (Jiss/Nils). Alternative: investigate BAdI 'Change Document Processing' as a cleaner approach

### 5 Verify change log recording

ME21N → create PO with Therapy Type = CELL\_GENE → Save ME22N → change Therapy Type to PLASMA → Save SE16N → CDPOS → filter CHANGENR for this PO Expected row: TABNAME=/ATO/S4\_PO\_HEADER\_EXT, FNAME=THERAPY\_TYPE, VALUE\_OLD=CELL\_GENE, VALUE\_NEW=PLASMA

■ OPEN ISSUE: ATO field changes ARE recorded in CDPOS correctly. However the ME22N/ME23N change log DISPLAY screen shows blank description for ATO rows. Joint call needed: ATO team (Allen Yuan) + PO team (Jiss) + Change Document team. This must be resolved before claiming GxP compliance for ATO field changes.

## 8

## Set Up BTP eSignature SaaS Communication Arrangement

*S4 ↔ BTP OAuth trust — enables S4 to call BTP eSignature service*

This Communication Arrangement is pre-wired when 84P is activated (Phase 2). If 84P was activated correctly, this may already be in place. Verify before setting up manually. Contact: Swarnava Chatterjee for BTP service details.

### 1 Verify if pre-wired by 84P activation

Transaction: /IWFND/MAINT\_SERVICE or Communication Arrangements app in S4HC Admin Search for a communication arrangement related to eSignature or GxP If found and active → skip to step 5 (verify connectivity)

### 2 Create Communication System in S4

S4HC Admin → Communication Systems → New System ID: BTP\_ESIGNATURE System Name: BTP eSignature SaaS Host Name: OAuth 2.0: Enabled Token URL: Client ID: Client Secret:

### 3 Get BTP service key credentials

BTP Cockpit → Subaccount → Instances → eSignature SaaS instance Create a Service Key if not existing Copy: clientid, clientsecret, url, tokenurl from the JSON service key

### 4 Create Communication Arrangement

S4HC Admin → Communication Arrangements → New Scenario: SAP\_COM\_XXXX (GxP eSignature — exact ID from SAP docs) Arrangement Name: CA\_BTP\_ESIGNATURE Communication System: BTP\_ESIGNATURE (from step 2) Authentication: OAuth 2.0 Client Credentials Save

### 5 Test connectivity

In Communication Arrangement → Test Connection button Expected: Green connection indicator — HTTP 200 from BTP eSignature endpoint If red: verify OAuth credentials, check BTP service is running, verify URLs

### 6 Verify CHECK\_ESIG\_ACTIVE returns TRUE

After Communication Arrangement is active AND 84P is activated: CHECK\_ESIG\_ACTIVE API should now return ABAP\_TRUE Test via ABAP Console or SE38 test report

■ Phase 8 complete when Communication Arrangement test = green AND CHECK\_ESIG\_ACTIVE = TRUE.

## Configure BTP — Register PO Object and Reason Codes

*BTP eSignature SaaS — define PURCHASE\_ORDER and its reason codes*

When 84P is activated, SAP-suggested reason codes are auto-provisioned in BTP. This phase covers verifying they are in place and adding custom ones if needed.

1

### Access BTP eSignature SaaS Admin

BTP Cockpit → Subaccount → eSignature SaaS application → Open Log in with admin credentials

2

### Verify PURCHASE\_ORDER object is registered

In eSignature Admin → Business Objects section PURCHASE\_ORDER should be listed (auto-provisioned by 84P activation) If not listed → create manually (step 3)

3

### Register PURCHASE\_ORDER manually if needed

Business Objects → Add New Object Name: PURCHASE\_ORDER Description: SAP S4 Purchase Order S4 System: Save

4

### Verify SAP-suggested reason codes are loaded

Select PURCHASE\_ORDER → Reason Codes tab SAP-suggested codes (auto-provisioned by 84P): • Clinical Protocol Update • Regulatory Submission • Error Correction • Therapy Type Reassignment

5

### Add custom reason codes if needed

Reason Codes → Add New Code: CUSTOM\_01 Description: Patient-Specific Order Change Save Note: Only reason codes assigned to PURCHASE\_ORDER appear in the PO reauthentication dropdown

6

### Test reason code dropdown

In ME21N → change Therapy Type → Save Reauthentication dialog should appear (only after Phase 10 is complete) Dropdown should show only PURCHASE\_ORDER reason codes

■ This phase is planned for 2702 and has NOT been implemented yet. This is the core eSignature integration work. Complete Phases 1-9 first.

## 1 Create BAdI implementation for ME\_PROCESS\_PO\_CUST

SE19 → BAdI: ME\_PROCESS\_PO\_CUST (Enhancement Spot: ME\_PURCHORD) Create implementation:  
ZCL\_PO\_ESIGN\_BADI\_IMPL Methods to implement: PROCESS\_HEADER, PROCESS\_ITEM

## 2 Implement PROCESS\_HEADER method

This fires when PO header is saved. Add reauthentication call:

```
METHOD if_ex_me_process_po_cust~process_header.

    DATA(lv_ebeln) = im_header->get_data( )-ebeln.

    IF lv_ebeln IS INITIAL. RETURN. ENDIF.

    " Step 1: Check if GxP is active in this system

    DATA lv_active TYPE abap_bool.

    " CALL METHOD cl_gxp_framework=>check_esig_active RECEIVING rv_active = lv_active.

    IF lv_active = abap_false. RETURN. ENDIF.

    " Step 2: Relevancy check – did any ATO regulated field change?

    DATA lv_relevant TYPE abap_bool.

    " Framework calls ZCL_PO_REAUTH_CHECK automatically via VC_SRA_SOT lookup

    IF lv_relevant = abap_false. RETURN. ENDIF.

    " Step 3: Trigger reauthentication dialog (LUW paused until user signs)

    " CALL METHOD cl_gxp_framework=>request_reauth

    "     EXPORTING iv_object = 'PURCHASE_ORDER'

    "             iv_ebeln = lv_ebeln.

    " Step 4: Validate result

    " CALL METHOD cl_gxp_framework=>validate_result

    "     RECEIVING rv_signed = DATA(lv_signed).

    " IF lv_signed = abap_false. RAISE ... ENDIF.
```

**3 Implement PROCESS\_ITEM method**

Same pattern as PROCESS\_HEADER but for item-level ATO fields Read im\_item->get\_data()-ebelp for the item number Check if THERAPY\_TYPE or PATIENT\_REF on the item changed

**4 Activate BAdI implementation**

SE19 → ZCL\_PO\_ESIGN\_BADI\_IMPL → Activate BAdI is now active — fires on every ME21N / ME22N save

**5 Test end-to-end in ME21N**

ME21N → create PO → enter TherapyType = CELL\_GENE → Save Expected: reauthentication dialog appears Enter password + select reason code → Confirm PO saved. CDHDR has new CHANGENR. BTP Signature ID linked.

**6 Verify complete audit trail**

SE16N → CDHDR → find PO → note CHANGENR SE16N → CDPOS → CHANGENR → THERAPY\_TYPE row with VALUE\_NEW=CELL\_GENE BTP eSignature Admin → Signature record linked to CHANGENR Combined = 21 CFR Part 11 compliant audit trail ✓

■ Phase 10 complete when reauthentication dialog appears in ME21N and CDHDR/BTP both have records.

# 11

## Set Up ATO SaaS — Event Mesh and iFlow

*BTP — enables bidirectional PO event exchange between ATO SaaS and S4*

### 1 Provision BTP Event Mesh

BTP Cockpit → Subaccount → Service Marketplace → Event Mesh → Create Instance Plan: Standard or Enterprise (based on event volume) Create service key → note connection details

### 2 Configure S4 outbound event binding

S4HC Admin → Enterprise Event Enablement Add topic binding for PO events: Topic: sap/s4/beh/purchaseorder/v1/PurchaseOrder/Changed/v1 Topic: sap/s4/beh/purchaseorder/v1/PurchaseOrder/Created/v1 Channel: your Event Mesh namespace Save → events will be published to Event Mesh on PO save

### 3 Deploy iFlow in Integration Suite

BTP → Integration Suite → Design → Import iFlow package from SAP API Hub: Package: ATO S4 Integration iFlows Contains: PO Create iFlow, PO Update iFlow, PO Cancel iFlow Configure: Source: Event Mesh topic subscription Target: S4 PurchaseOrder OData V4 API endpoint Auth: OAuth credentials from Phase 8 Communication Arrangement

### 4 Configure iFlow schema mapping

ATO SaaS event schema ≠ S4 OData API schema iFlow mapping handles: ATO TherapyId → S4 TherapyType ATO StudyRef → S4 ClinicalStudy ATO ProtocolId → S4 ProtocolRef ATO PatientId → S4 PatientRef Deploy and activate iFlow

### 5 Configure ATO SaaS event subscription

ATO Cloud Admin → Integration Settings S4 Endpoint: your S4 OData V4 PurchaseOrder API URL Event Mesh: connection details from step 1 Subscribe to topics: PurchaseOrder.Created, PurchaseOrder.Changed

### 6 Test event flow

ATO SaaS → trigger PO creation (Manage Schedules → Trigger ERP Document Creation) Verify: BTP Event Mesh → message published Integration Suite Monitor → iFlow COMPLETED S4 ME23N → PO created with ATO fields S4 → outbound event published back to ATO ATO portal → PO number linked to specimen shipment

■ Phase 11 complete when ATO triggers PO in S4 and S4 notifies ATO of changes.

## End-to-End Validation Checklist

Run after all phases complete — covers ATO → S4 → eSign → Reauth

| #       | Validation                    | How                                                    | Expected Result                |
|---------|-------------------------------|--------------------------------------------------------|--------------------------------|
| ■<br>1  | Add-on installed              | SAINT → LS_ADD_ON                                      | Status = INSTALLED             |
| ■<br>2  | 84P activated                 | SPRO → GxP node visible,<br>CHECK_ESIG_ACTIVE=TRUE     | Both conditions met            |
| ■<br>3  | VC_SRA_SOT configured         | SE16N → VC_SRA_SOT →<br>PURCHASE_ORDER row             | Object + class names present   |
| ■<br>4  | Callback class active         | SE24 → ZCL_PO_REAUTH_CHECK →<br>Active (blue)          | Methods implemented and active |
| ■<br>5  | OData V4 extensibility        | GET /PurchaseOrder/\$metadata →<br>TherapyType visible | ATO properties in metadata     |
| ■<br>6  | ATO tables have data          | SE16N → /ATO/S4_PO_HEADER_EXT after<br>PO creation     | Row with ATO field values      |
| ■<br>7  | Change Doc records ATO fields | SE16N → CDPOS →<br>TABNAME=/ATO/S4_PO_HEADER_EXT       | Rows with VALUE_OLD/VALUE_NEW  |
| ■<br>8  | Comm Arrangement active       | Comm Arrangement → Test Connection →<br>green          | HTTP 200 from BTP              |
| ■<br>9  | BTP object registered         | BTP eSign Admin → PURCHASE_ORDER<br>listed             | Object with reason codes       |
| ■<br>10 | Reauth dialog in ME21N        | ME21N → change Therapy Type → Save →<br>dialog appears | Password + reason code shown   |
| ■<br>11 | Signature linked to CHANGENR  | CDHDR CHANGENR matches BTP<br>Signature ID             | End-to-end audit trail         |
| ■<br>12 | ATO event creates PO          | ATO portal → trigger → PO created in S4                | ME23N shows PO with ATO tab    |
| ■<br>13 | S4 notifies ATO of change     | ME22N change → Event Mesh → ATO portal<br>updated      | ATO shows changed value        |
| ■<br>14 | Cancel flow works             | ATO cancel → S4 PO item deletion indicator             | ME23N item shows LOEKZ=X       |

### Key Contacts for Each Phase

| Phase | Area                        | Contact                      | Email                       |
|-------|-----------------------------|------------------------------|-----------------------------|
| 1     | Life Science Add-on install | SAP Cloud Operations         | Via support ticket          |
| 2     | 84P activation              | SAP Cloud Operations / Basis | Via support ticket          |
| 3     | VC_SRA_SOT config           | Yadesh Gupta (Framework)     | ya.gupta@sap.com            |
| 4     | Callback class dev          | PO LOB development team      | Nils Hartmann               |
| 5     | OData V4 API extensibility  | Nils Hartmann / Jiss         | PO standard team            |
| 6     | ATO extension tables        | Loring Wu / Allen Yuan       | ATO dev team                |
| 7     | Change Doc + EINKBELEG      | Allen Yuan + Jiss            | ATO + PO standard           |
| 8     | Comm Arrangement / BTP      | Swarnava Chatterjee          | swarnava.chatterjee@sap.com |
| 9     | BTP reason codes            | Swarnava Chatterjee / Yadesh | eSignature + framework team |
| 10    | ME21N BAdI integration      | PO LOB dev team              | Nils Hartmann               |

|    |                    |                               |                      |
|----|--------------------|-------------------------------|----------------------|
| 11 | Event Mesh / iFlow | Loring Wu / Viswanath Natesan | ATO integration team |
|----|--------------------|-------------------------------|----------------------|